



END SEMESTER ASSESSMENT (ESA) - Jan - May 2024

UE19CS353 - Object Oriented Analysis and Design with Java

Total Marks : 100.0

1.a. Briefly describe the need of garbage collector in java.

(4.0 Marks)

1.b. **What is the need for constructors in Java? With an example java program explain default and parameterized constructors** (6.0 Marks)

1.c. **Briefly describe Abstraction, Inheritance and Polymorphism concepts of Object oriented programming** (6.0 Marks)

1.d. Explain any two different types of inheritance in java with an example for each. (4.0 Marks)

2.a. Describe in a line or two the following elements of component diagram: Port, Dependency, Component, Interface. (4.0 Marks)

2.b. Draw a use case diagram for ATM machine (6.0 Marks)

2.c. What are Generic classes in java. Write an example program to demonstrate generic class in java. (6.0 Marks)

2.d. Explain in a line or two the following terms with respect to a class diagram

Association

Aggregation

Composition

Qualified Association

(4.0 Marks)

3.a. Draw an activity diagram to login and logout for a web based application

(6.0 Marks)

3.b. With respect to state diagram describe the following terms in line or two and give an example for each.

State

Event

Guard condition

(6.0 Marks)

3.c. Explain in a line or two the following GRASP principles with an example.

i) Creator

ii) Low Coupling

iii) Controller

iv) Polymorphism

(8.0 Marks)

4.a. Describe any four solid design principles with an example for each. (8.0 Marks)

4.b. What is a singleton class. How to make a class singleton (6.0 Marks)

4.c. What are Design patterns? Which are the different types of design patterns, describe in a line or two? Specify pros and cons of design pattern (6.0 Marks)

5.a. Give the UML diagram of the Adapter pattern and briefly describe each component (8.0 Marks)

5.b. Briefly describe the intent, structure and participants of chain of responsibility design pattern. (7.0 Marks)

5.c. What is an Anti pattern? Briefly describe Lava-Flow anti pattern (5.0 Marks)